Arithmetic Vacuum Engineering

From Bost-Connes Phase Transitions to Warp Drive via Sheaf Cohomology on $\mathrm{Spec}(\mathbb{Z})$

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A monograph in three parts:

Part I. Mechanism and Experiment
Part II. Theoretical Chain and Open Problems
Part III. Sheaf-Theoretic Reformulation

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Overview

This monograph presents a research program connecting the arithmetic structure of the Riemann zeta function to the possibility of engineering negative vacuum energy density — the principal requirement for Alcubierre-type warp metrics.

The central observation is that the quantum vacuum energy, computed via zeta function regularization, inherits the Euler product decomposition of $\zeta(s) = \prod_p (1-p^{-s})^{-1}$. Each prime p contributes an independent "channel" to the vacuum. Suppressing the channel at prime p — an operation we call *prime channel muting* — modifies the vacuum energy by the factor $(1-p^{-s})$, which is negative at the physically relevant point $s = -3$.

Part I develops the mechanism (via the Bost-Connes quantum statistical mechanical system), proves the sign-flip theorem, identifies three physical implementation paths, and proposes a concrete experiment using superconducting circuits.

Part II places the mechanism in a ten-step theoretical chain from the Bost-Connes system to warp drive, classifies each step as established, argued, or open, and formulates five sub-problems for the remaining scaling challenge.

Part III reformulates the entire framework in the language of sheaves and cohomology on $\mathrm{Spec}(\mathbb{Z})$, proposes that the weak energy condition is a discrete topological invariant (regulator orientation), and analyzes the physics of the boundary between "normal" and "muted" vacuum regions.

Part I

Mechanism and Experiment

1 Introduction

The Alcubierre warp metric [1] permits superluminal travel within general relativity but requires matter violating the weak energy condition (WEC): negative energy density. The Casimir effect [2] demonstrates that boundary conditions on quantum fields can produce negative vacuum energy, but the magnitude falls short by many orders of magnitude [3].

The central observation of this work is that the quantum vacuum has *arithmetic structure*: its energy is computed via zeta function regularization, and $\zeta(s)$ decomposes as an Euler product over primes. Each prime contributes an independent channel. We show that suppressing a single channel flips the sign of vacuum energy.

2 Zeta-Regularized Vacuum Energy

2.1 Casimir energy and zeta regularization

The zero-point energy of a quantum field with modes $\{\omega_n = n\omega_0\}$ is regularized as

$$E_{\text{vac}}^{\text{reg}} = \frac{\hbar\omega_0}{2}\,\zeta_{\mathcal{H}}(-d), \tag{1}$$

where d is the spatial dimension and $\zeta_{\mathcal{H}}(s) = \zeta(s)$ for a standard cavity. In three dimensions:

$$E_{\text{vac}}^{(3D)} = \frac{\hbar\omega_0}{2}\,\zeta(-3) = \frac{\hbar\omega_0}{240}. \tag{2}$$

2.2 Euler product and prime channels

The Euler product

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1-p^{-s}} \tag{3}$$

factorizes $\zeta(s)$ into independent prime contributions.

Definition (Prime channel muting). The p-muted zeta function is

$$\zeta_{\neg p}(s) \equiv \zeta(s)\cdot(1-p^{-s}) = \prod_{q\neq p}\frac{1}{1-q^{-s}} = \sum_{\substack{n=1\\ p\nmid n}}^{\infty} n^{-s}. \tag{4}$$

Geometrically, $\zeta_{\neg p}(s)$ is the zeta function of the open subscheme $\mathrm{Spec}(\mathbb{Z}[1/p]) = \mathrm{Spec}(\mathbb{Z})\setminus\{(p)\}$.

3 The Bost-Connes System

The Bost-Connes (BC) system [4, 5] acts on $\ell^2(\mathbb{N}^*)$ with $H|n\rangle = \log(n)|n\rangle$ and partition function $Z(\beta) = \zeta(\beta)$. It exhibits a phase transition at $\beta = 1$ with symmetry group $\mathrm{Gal}(\mathbb{Q}^{\text{ab}}/\mathbb{Q})$.

Definition (Prime projection). $P_{\neg p} = \sum_{p \nmid n} |n\rangle\langle n|$ projects onto $\mathcal{H}_{\neg p} = \mathrm{span}\{|n\rangle : \gcd(n,p) = 1\}$. The restricted partition function is

$$Z_{\neg p}(\beta) = \mathrm{Tr}_{\mathcal{H}_{\neg p}}(e^{-\beta H}) = \zeta(\beta)(1 - p^{-\beta}). \tag{5}$$

This is exact (an algebraic identity from the Euler product).

Three physical mechanisms realize $P_{\neg p}$: **(A)** spectral filtering (notch filter at $\omega = \log p$); **(B)** arithmetic boundary conditions (periodic structure with period p); **(C)** phase transition sector selection (freeze $\mathbb{Z}_p^*$ component of Galois action).

4 The Sign-Flip Theorem

Theorem 4.1 (Vacuum energy sign flip)**.** *For any prime $p \geq 2$ and positive integer n,*

$$\zeta_{\neg p}(1 - 2n) = \zeta(1 - 2n) \cdot (1 - p^{2n-1}). \tag{6}$$

Since $p^{2n-1} \geq 2 > 1$, the factor $(1 - p^{2n-1})$ is strictly negative. Therefore $\zeta_{\neg p}(1 - 2n)$ and $\zeta(1 - 2n)$ have opposite signs.

Proof. Direct evaluation of $\zeta_{\neg p}(s) = \zeta(s)(1 - p^{-s})$ at $s = 1 - 2n$. □

Corollary 4.2. *The 3D vacuum energy under p-muting is $\zeta_{\neg p}(-3) = (1 - p^3)/120 < 0$ for all $p \geq 2$.*

Table 1: Sign-flip for the first primes ($s = -3$).

p	$(1 - p^3)$	$\zeta_{\neg p}(-3)$
2	-7	$-7/120$
3	-26	$-26/120$
5	-124	$-124/120$
7	-342	$-342/120$

5 Proposed Experiment

We propose a $p = 2$ prime filter using a Nb coplanar waveguide resonator ($L = 10\,\mathrm{mm}$, $f_0 \approx 6.0\,\mathrm{GHz}$) with ten SQUID notch filters tuned to even harmonics $2f_0$ through $20f_0$. At base temperature $T < 20\,\mathrm{mK}$ (dilution refrigerator), thermal occupation is negligible ($\langle n_{\mathrm{th}} \rangle \sim 10^{-9}$).

Protocol: Compare vacuum energy with all SQUIDs detuned (Config. A) vs. tuned to even harmonics (Config. B).

Prediction: Vacuum energy sign flip. $E_{\mathrm{odd}}/E_{\mathrm{full}} \to \zeta_{\neg 2}(-1)/\zeta(-1) = -1$.

Success criteria: Level 1: $\Delta E \neq 0$; Level 2: matches ζ-regularized value; Level 3: multiplicative structure for $p = 2, 3$; Level 4: controllable sign-flipping.

Resources: Dilution refrigerator lab, \$50–100K, 12 months.

6 Implications for Warp Drive

The Alcubierre metric requires $\rho < 0$ in the bubble wall. Theorem 4.1 produces exactly this. However, the connection from circuit-scale to macroscopic spacetime requires bridging six steps, of which only the first three are experimentally testable by our proposal. The energy gap of $\sim 10^{69}$ orders of magnitude (the scaling problem) remains open.

Part II

Theoretical Chain and Open Problems

7 The Ten-Step Chain

Table 2: Ten-step chain. E = Established, A = Argued, O = Open.

#	Step	Status	Key reference
1	NCG Standard Model ($M^4 \times F$)	E	Connes [6]
2	BC system = arithmetic kernel of F	A	Connes-Marcolli [5]
3	Phase transition at $\beta = 1$	E	Bost-Connes [4]
4	Prime muting ($P_{\neg p}$)	E	Part I
5	Partition function identity	E	Euler product
6	Vacuum energy sign flip	E	Theorem 4.1
7	WEC violation	E	Standard QFT
8	Warp metric allowed	E	Alcubierre [1]
9	NCG $\leftrightarrow$ bulk spacetime	A	[5, 9]
10	Scaling ($10^{-22} \to 10^{47}$ J)	O	—

8 Argued Steps: Precise Conjectures

Conjecture 8.1 (Arithmetic kernel)**.** *The internal space F in $M^4 \times F$ contains $\mathcal{A}_{\mathrm{BC}} = C^*(\mathbb{Q}_+^* \backslash \mathbb{A}_f / \hat{\mathbb{Z}}^*)$ as an arithmetic subalgebra. KMS states of $\mathcal{A}_{\mathrm{BC}}$ correspond to vacuum sectors of quantum fields on M^4.*

Evidence: (i) Both F and $\mathcal{A}_{\mathrm{BC}}$ are NC C^*-algebras with Galois-type automorphisms; (ii) the spectral action on $M^4 \times F$ produces ζ-function values, and the BC partition function *is* ζ; (iii) Connes-Marcolli [5], Theorem 4.1, establishes the BC algebra as the NC space of $\mathbb{Q}$-lattices. **Missing**: explicit embedding $\mathcal{A}_{\mathrm{BC}} \hookrightarrow \mathcal{A}_F$.

Step 9 (NCG $\leftrightarrow$ bulk curvature) has two candidate bridges: (A) spectral action modification, (B) holographic (AdS/CFT-type). Both are conjectural.

9 The Scaling Problem: Five Sub-Problems

I. Coherent amplification: N synchronized systems $\Rightarrow E \propto N^2$? Under what conditions? Decoherence rate?

II. Topological protection: $K_1(\mathbb{Z}) = \mathbb{Z}/2 \to K_1(\mathbb{Z}[1/p]) = \mathbb{Z}/2 \times \mathbb{Z}$ is a topological phase transition. Is it physically realizable?

III. Critical amplification: Critical exponents of BC system at $\beta = 1$? Does prime muting couple to the critical mode?

IV. Holographic amplification: Arithmetic AdS/CFT? Central charge?

V. Bootstrap: Planck-scale bubble stability? Energy cost vs. radius?

10 Roadmap

Phase 1 (1–2 yr): SQUID experiment (Steps 4–6). Phase 2 (2–5 yr): Sheaf theory + topological phases (Part III). Phase 3 (5–10 yr): Coherent arrays + critical phenomena. Phase 4 (10+ yr): Macroscopic negative energy, if Phases 1–3 succeed.

Part III

Sheaf-Theoretic Reformulation

11 Motivation: Cantor vs. Grothendieck

Part I treats $\zeta(s) = \sum n^{-s}$ as a sum over points (set-theoretic, Cantor). Part III treats it as the zeta function of the scheme $\mathrm{Spec}(\mathbb{Z})$ (sheaf-theoretic, Grothendieck).

The difference: *set-theoretic muting* removes terms from a sum (quantitative); *sheaf-theoretic muting* changes the underlying space via localization $\mathbb{Z} \to \mathbb{Z}[1/p]$ (qualitative).

12 The Energy Sheaf on $\mathrm{Spec}(\mathbb{Z})$

Proposition 12.1 (Energy sheaf)**.** *The assignment $\mathcal{F}(D(n)) = \zeta(s) \cdot \prod_{p|n}(1 - p^{-s})$ with restriction maps $f \mapsto f \cdot \prod_{p|n,\, p\nmid m}(1 - p^{-s})$ defines a sheaf on* $\mathrm{Spec}(\mathbb{Z})$ *(Zariski topology).*

Global sections: $\mathcal{F}(\mathrm{Spec}(\mathbb{Z})) = \zeta(s)$. Sections over $D(p)$: $\mathcal{F}(D(p)) = \zeta_{\neg p}(s)$.

13 Localization Exact Sequence

For the multiplicative group sheaf $\mathbb{G}_m$ on $\mathrm{Spec}(\mathbb{Z})_{\text{ét}}$:

$$0 \to \underbrace{\mathbb{Q}_p/\mathbb{Z}_p}_{H^2_{(p)}} \to \mathrm{Br}(\mathbb{Z}) \to \mathrm{Br}(\mathbb{Z}[1/p]) \to 0. \tag{7}$$

Localization at p removes $\mathbb{Q}_p/\mathbb{Z}_p$ (all p-local obstructions) from the Brauer group.

14 K-Theory and the Regulator

$K_1(\mathbb{Z}) = \mathbb{Z}/2$, $K_1(\mathbb{Z}[1/p]) = \mathbb{Z}/2 \times \mathbb{Z}$ (free rank increases by 1 per prime muted). $K_2(\mathbb{Z}) = \mathbb{Z}/2$ with generator $\{-1, -1\}$ (Milnor).

The Borel regulator $\mathrm{reg} : K_{2n-1}(\mathbb{Z}) \to \mathbb{R}^{d_n}$ maps to a lattice whose covolume involves $\zeta(n)$ [12, 13].

Theorem 14.1 (Regulator sign flip)**.** *For any prime $p \geq 2$ and positive integer n, the localized regulator value $\zeta_{\neg p}(1 - 2n) = \zeta(1 - 2n) \cdot (1 - p^{2n-1})$ has opposite sign from $\zeta(1 - 2n)$.*

Numerically identical to Theorem 4.1, but the K-theoretic interpretation is new:

Conjecture 14.2 (Regulator orientation)**.** *The sign of $\zeta(1 - 2n)$ is a topological invariant: the* orientation *of the Borel regulator lattice. Localization reverses this orientation. Physically, this determines whether the WEC is satisfied (positive) or violated (negative).*

If correct, WEC is a discrete (on/off) invariant, not a continuous energy threshold. The scaling problem dissolves.

15 Five Physical Interpretations of Localization

$\mathbb{Z} \to \mathbb{Z}[1/p]$ admits five equivalent physical readings:

1. **Gauging**: $|n\rangle \sim |pn\rangle$ identified (discrete gauge symmetry).
2. **Kaluza-Klein**: p-adic extra dimension decompactified ($R_p \to \infty$).
3. **Phase transition**: p's order parameter frozen (selective $\beta_p \to \infty$).
4. **Topos logic**: Heyting algebra changes; WEC truth value flips.
5. **Adelic**: $\mathbb{Q}_p$ component removed from $\mathbb{A}_\mathbb{Q}$.

Most experimentally accessible: (1) discrete gauge theories in qubit arrays; (3) phase transitions in quantum simulators; and the K-theoretic topological phase transition $K_1(\mathbb{Z}) \to K_1(\mathbb{Z}[1/p])$ realized in condensed matter.

16 Boundary Physics: The Arithmetic Domain Wall

If localization is applied to a finite region, the boundary $V(p) = \mathrm{Spec}(\mathbb{F}_p)$ separates WEC-true from WEC-false domains.

Local cohomology $H^2_{(p)} \cong \mathbb{Q}_p/\mathbb{Z}_p$ lives on the wall: an infinite tower of "arithmetic edge states" with p^n modes at resolution level n.

Predicted wall phenomena: (i) topological edge states (from $\Delta K_1 = \mathbb{Z}$); (ii) p-adic radiation at $T_{\text{wall}} \sim \hbar\omega_0 \log(p)/(2\pi k_B)$; (iii) refractive index discontinuity $n_{\text{in}}/n_{\text{out}} = \sqrt{1 - 1/p}$; (iv) topological stability via tame ramification (wall minimum thickness $\sim \ell_P \log p$).

This arithmetic domain wall is the warp bubble wall: a closed surface in space separating $\mathrm{Spec}(\mathbb{Z})$ (normal vacuum) from $\mathrm{Spec}(\mathbb{Z}[1/p])$ (WEC-violating vacuum), with the Alcubierre shape function encoded in the transition of the structure sheaf.

17 What Remains Conjectural

1. Conjecture 14.2 (WEC = regulator orientation) is unproven.
2. Wall phenomena assume that mathematical bulk-boundary correspondence has physical content.
3. The dissolution of the scaling problem depends entirely on Conjecture 14.2. If WEC is fundamentally a continuous condition, scaling remains.

Conclusion

This monograph presents a three-level analysis of arithmetic vacuum engineering:

Level 1 (Part I): An algebraically exact mechanism (prime channel muting via the Bost-Connes system) that flips the sign of zeta-regularized vacuum energy, with a concrete and affordable experimental proposal.

Level 2 (Part II): A ten-step theoretical chain connecting this mechanism to warp drive, with seven steps established, two argued via precise conjectures, and one open (scaling).

Level 3 (Part III): A sheaf-theoretic reformulation that reinterprets the sign flip as a regulator orientation reversal on $\mathrm{Spec}(\mathbb{Z})$, potentially dissolving the scaling problem by recasting it as a topological phase transition rather than an energy accumulation challenge.

The key experimental test — the SQUID prime filter experiment — can be conducted with existing technology for \$50–100K in 12 months. Success at Level 3 (multiplicative structure confirmed for multiple primes) would establish that the Euler product formula is a physical law governing the quantum vacuum, independent of its implications for warp drive.

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